

Lecture 10: Statistics continued

CMSC/DATA 320: Introduction to Data Science

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Announcements

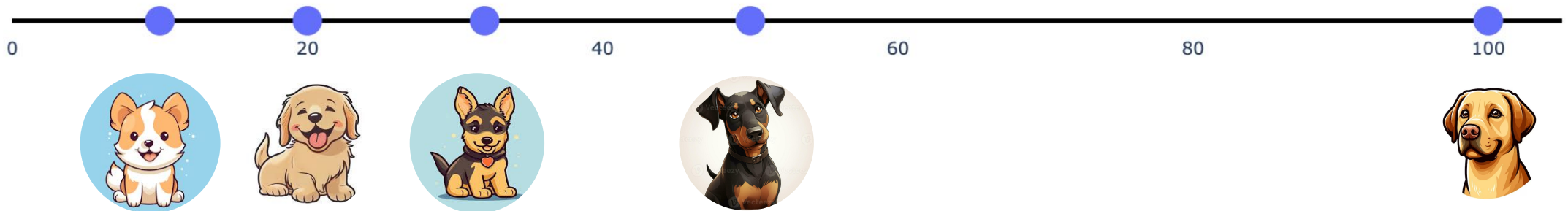
- PA3, WA3 came out yesterday
 - Due March 8
 - Recommendation: take a look at them this week
- 

Central Limit Theorem



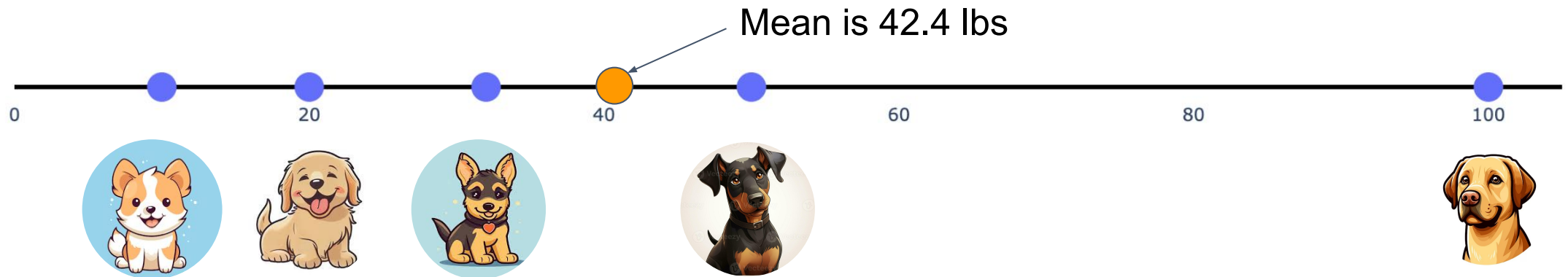
Motivating Example

Example: Say we wanted to know the average weight of dogs. But we could only measure 5 dogs' weight. So say we weighed 5 dogs



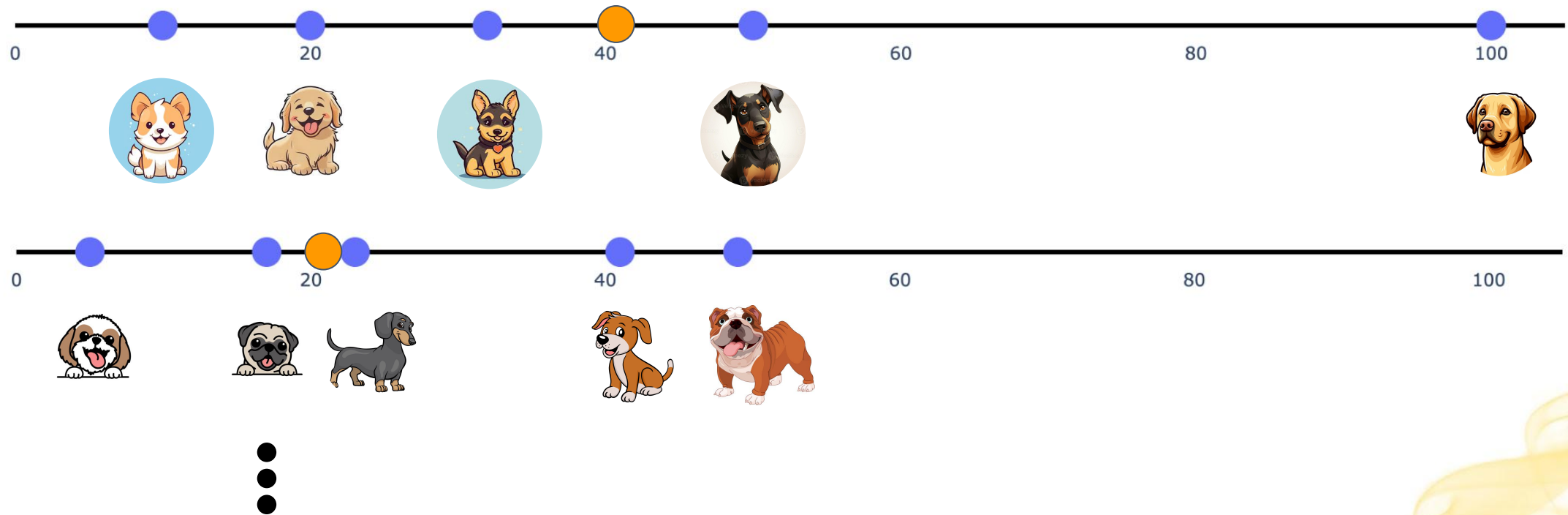
Standard Error vs Standard Deviation

Example: Say we weighed 5 dogs



Standard Error vs Standard Deviation

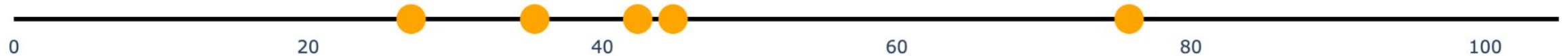
Now suppose we did this 5 times with different dogs (25 dogs total)



We get 5 different sample means

Distribution of sample means

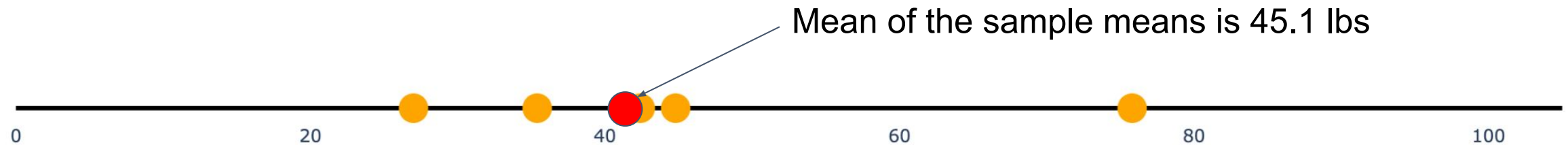
If we plot those 5 sample means on a line, we have



If we have many such means and plot them on a histogram, they will be distributed normally.

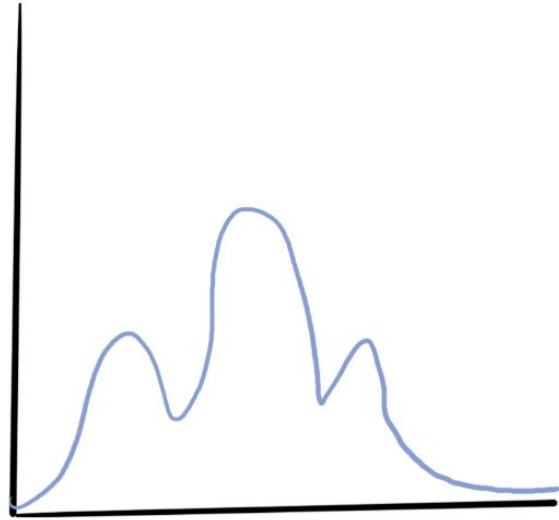
Distribution of sample means

We can take the mean of those 5 sample means

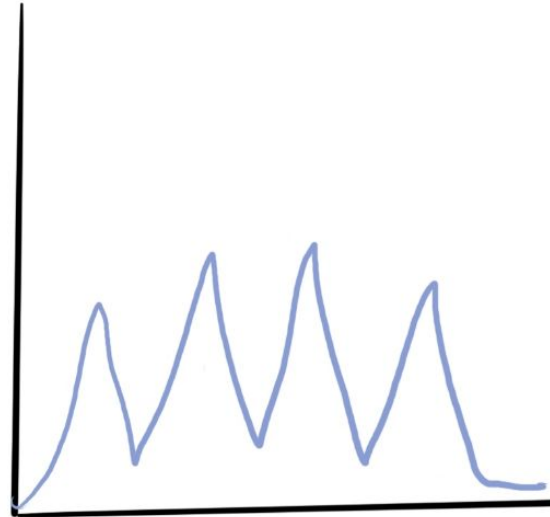


If we have many such means (orange points), the mean of the means (red point) will be the true population mean.

Comparing two non-normal distributions is hard.



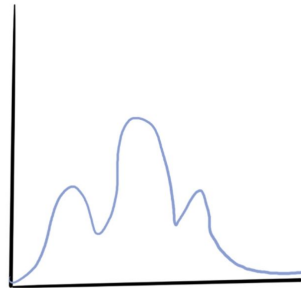
Some annoying distribution



Some other, even more annoying distribution

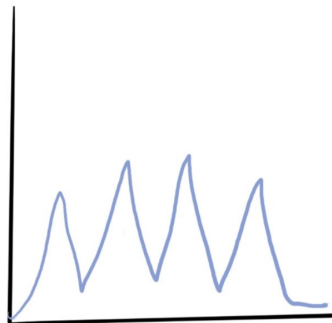
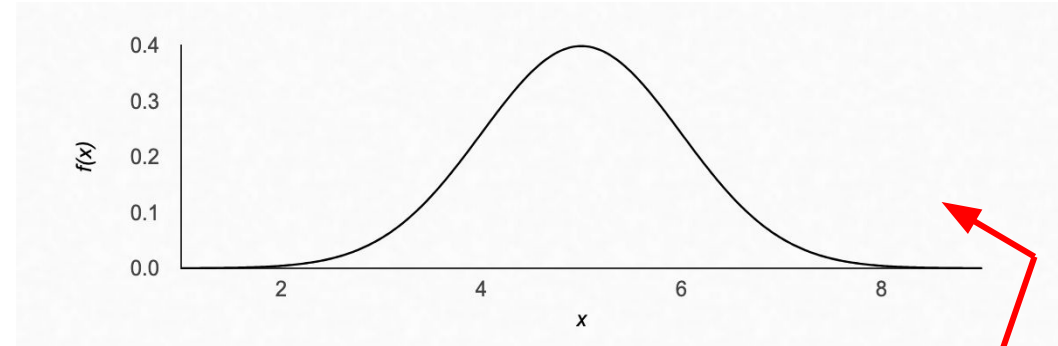
Sampling from non-normal distributions

The two
distributions



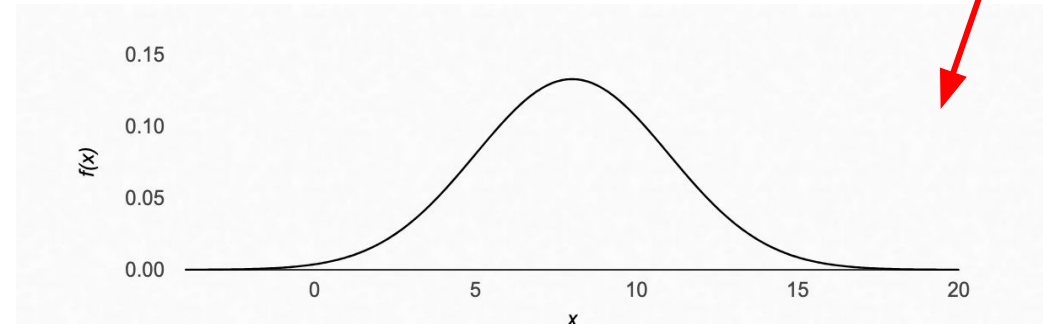
Some annoying distribution

Take a bunch of
samples



Some other, even more
annoying distribution

Take a bunch of
samples



We took a bunch
of samples, took
the mean of each,
and plotted
the histograms
for each original
distribution

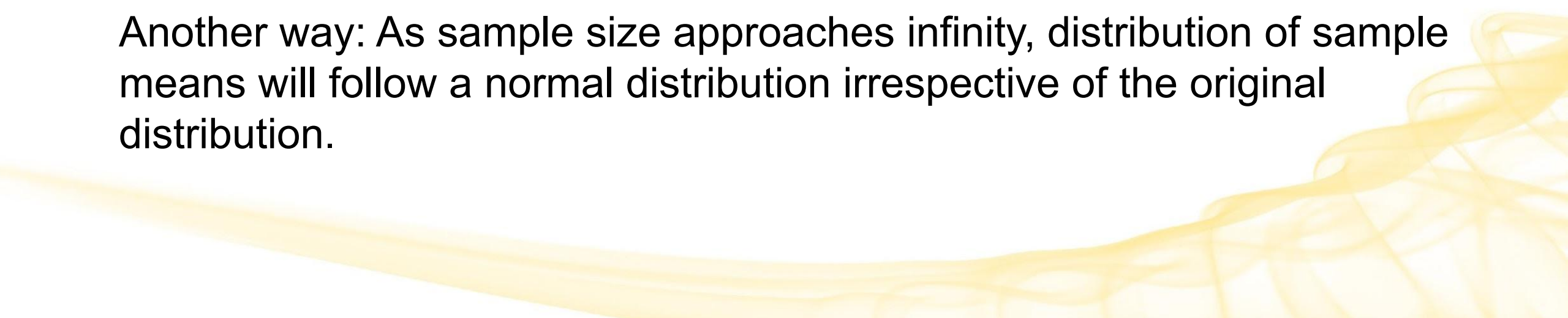
Central Limit Theorem

One of the fundamental theorems in statistics – core to statistical analysis!

States:

“Given a sufficiently large sample size, the sampling distribution of the mean of a variable will approximate a normal distribution regardless of that variable’s distribution in a population.”

Another way: As sample size approaches infinity, distribution of sample means will follow a normal distribution irrespective of the original distribution.



How many is “sufficiently large”?

CLT: “Given a **sufficiently large** sample size, the sampling distribution of the mean of a variable will approximate a normal distribution regardless of that variable’s distribution in a population.”

A sample size is considered "large" if it's at least 30

$n \geq 30 \rightarrow$ “large enough”

However

- Keep in mind that CLT is about a *sampling distribution* but in practice we’d be getting a finite number of samples

And

- If the population is **normal**, then the result holds for samples of any size
 - The sampling distribution of the sample means will be approximately normal even for samples of size less than 30!

Example: Rolling a die



Sampling from a Uniform Distribution

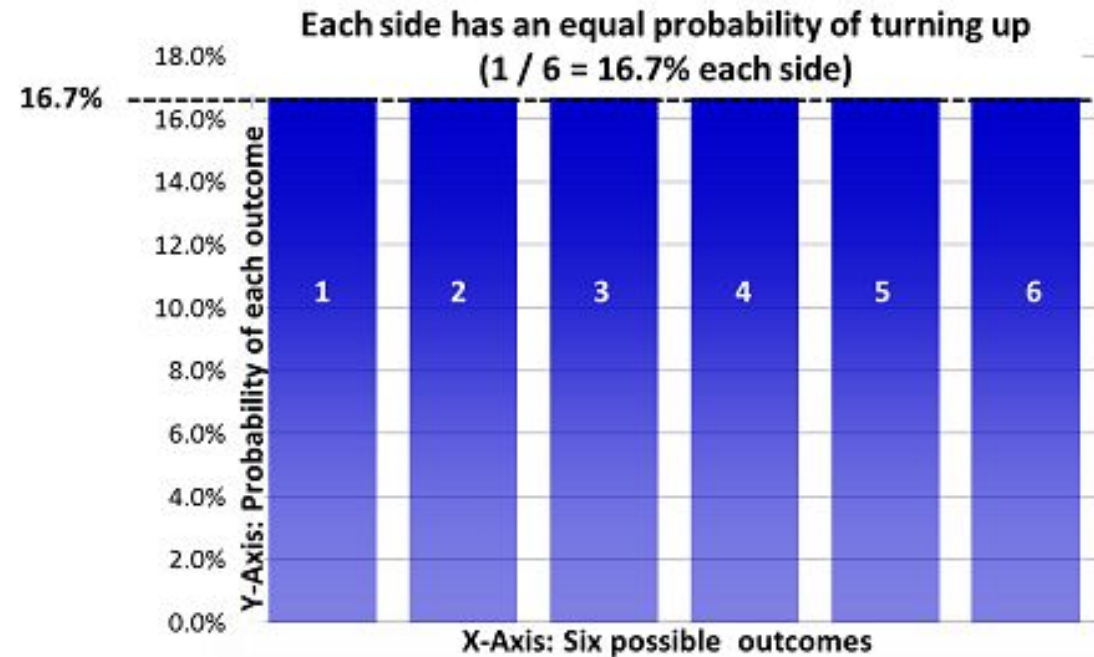
Scenario:

Roll the die 5 times – that's one sample.

Most of the time, the mean will be 3.5

Sometimes, it won't be!

What is the rarest mean?



Example: Back to rolling a die 🎲

If we roll a die several times we might get:

1 2 3 4 5 → Average of 3

1 1 6 6 3 → Average of 3.4

1 1 1 1 1 → Average of 1

6 6 6 6 6 → Average of 6

We know we have the highest chances of getting 3.5.

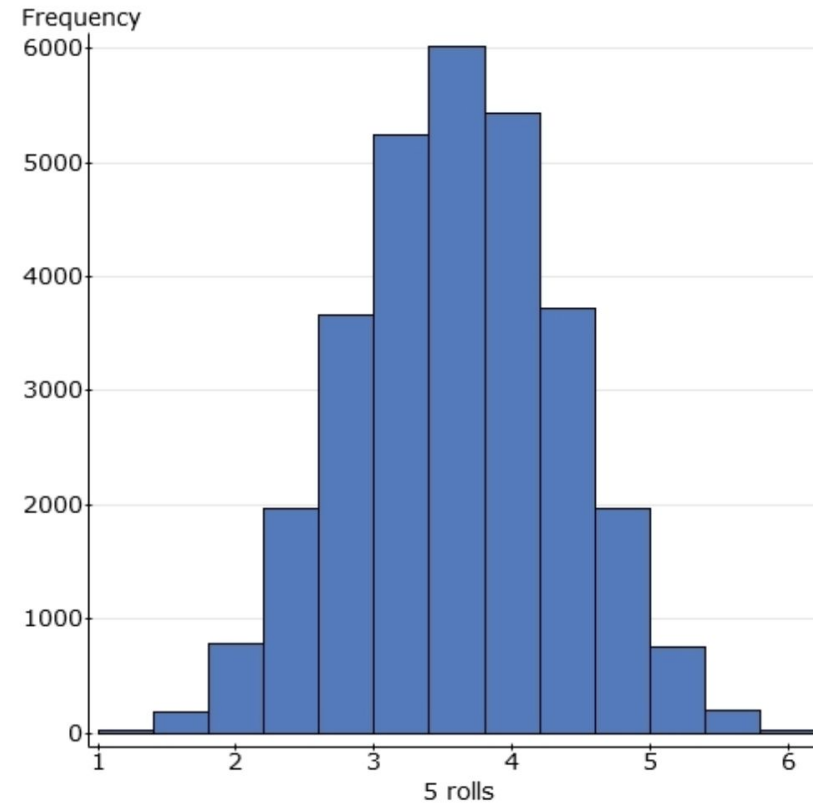
Is there a way we can discuss the likelihood of getting the other means?

Yes. A distribution.



Sampling

If you roll a die five times, here is the frequency with which you will get each of the possible means.

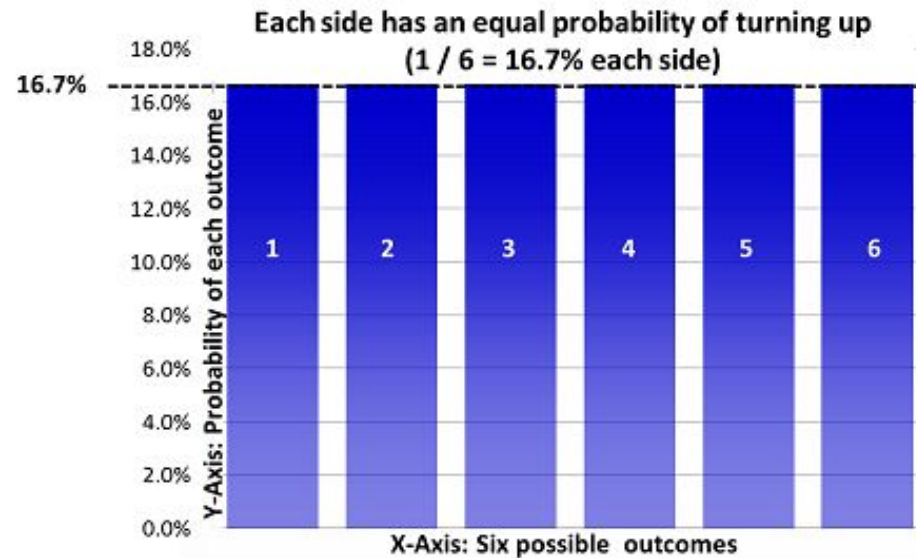


Histogram showing the frequency of each possible mean when rolling a die five times

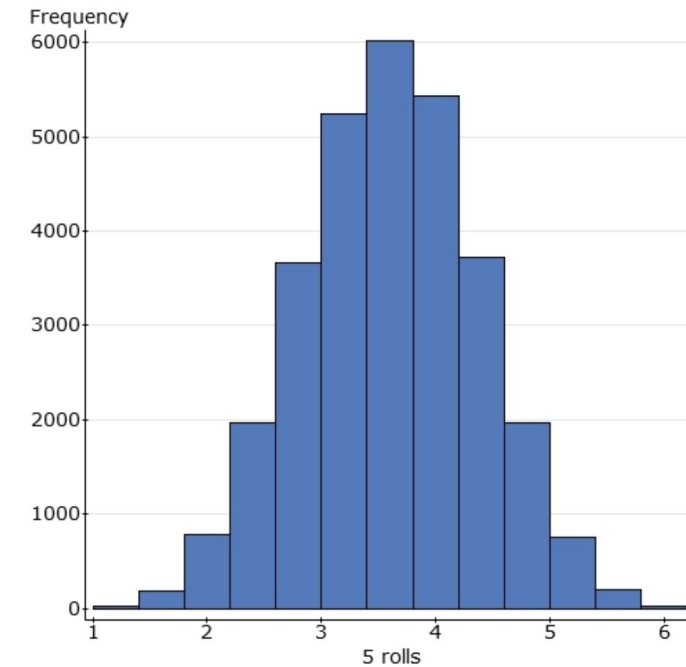
Sampling

According to the Central Limit Theorem, if you roll a die many times and calculate the mean each time, **the sample means will follow a normal distribution.**

We now have two separate things:



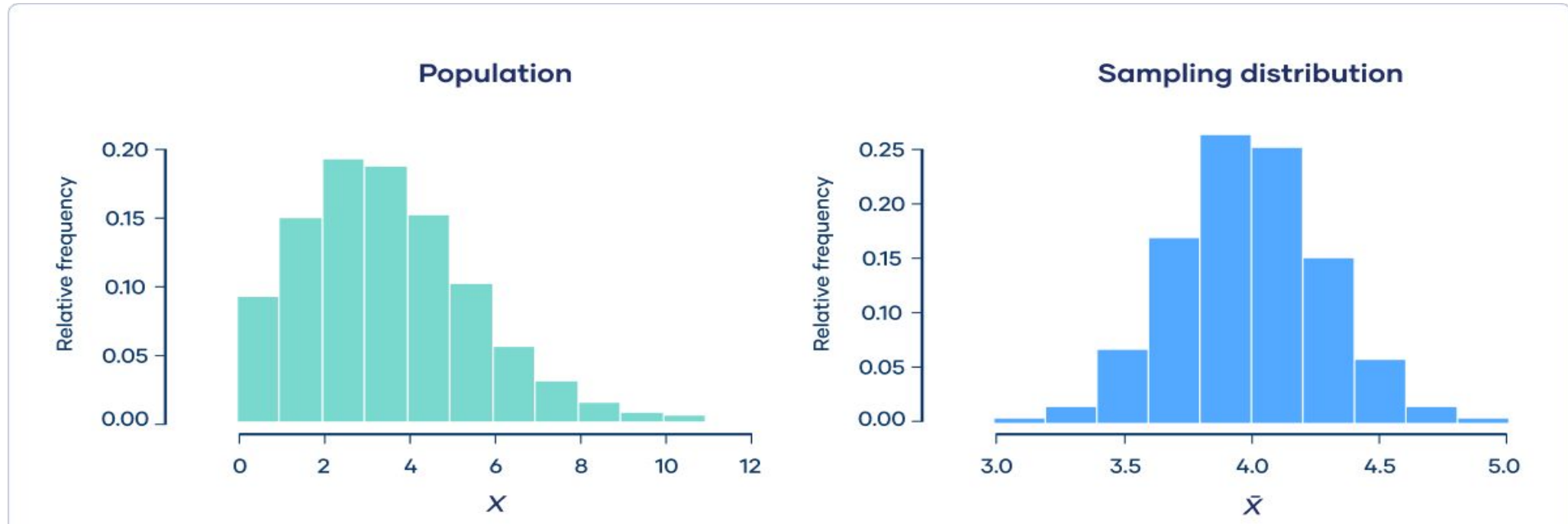
The original distribution



The potential distribution of means when we sample it

Inference: We started with the uniform data distribution, but **the means of samples** drawn from it resulted in a normal distribution.

Another Example: The Central Limit Theorem



A **population** follows a **Poisson distribution** (left image). If we take 10,000 **samples** from the population, each with a sample size of 50, the sample means follow a normal distribution, as predicted by the **central limit theorem** (right image).

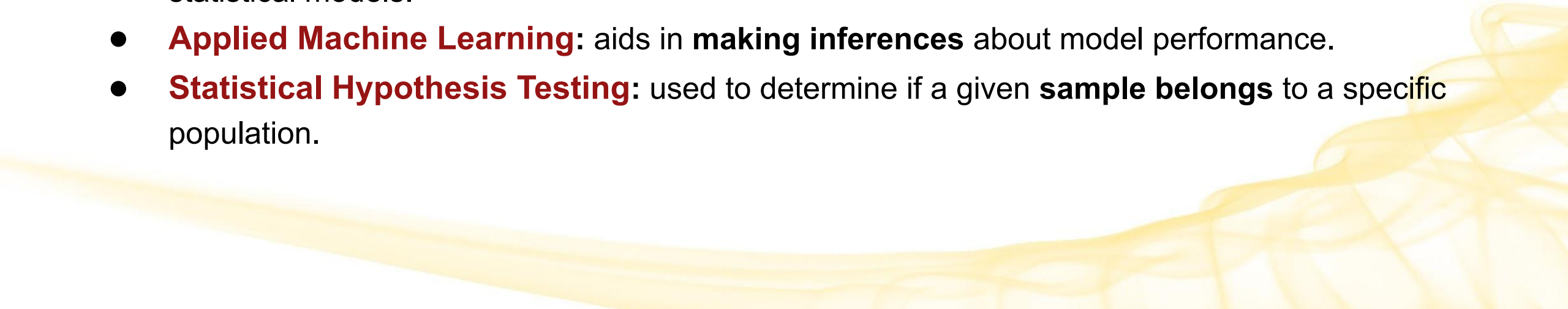


Central Limit Theorem Application

The CLT is useful in various scenarios, particularly when analyzing an entire population is difficult.

- It allows us to assume that the sampling distribution of the mean will be normally-distributed in most cases

Some key applications:

- **Data Science**: helps make **accurate assumptions** about a population to build robust statistical models.
 - **Applied Machine Learning**: aids in **making inferences** about model performance.
 - **Statistical Hypothesis Testing**: used to determine if a given **sample belongs** to a specific population.
- 

Standard Error (SE)

The **Standard Error** of a sample statistic (ex. mean, median) is the **standard deviation** of that statistic's **sampling distribution**, given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

n – size of each sample!!

If the statistic is the sample **mean**, it is called the **Standard Error of the Mean** (SEM).

- SE and SEM are often used interchangeably, though strictly speaking SE could be any sample statistic.
- Mean is most commonly used

If we only have one sample with mean s , we'll estimate the SE like so:

$$SE \approx \frac{s}{\sqrt{n}}$$

Standard Error vs Standard Deviation

Standard Deviation gives us the variability of values in a sample of the population

ex. how far a data point is likely from the mean

Standard Error gives us the variability of a statistic between samples drawn from the same population

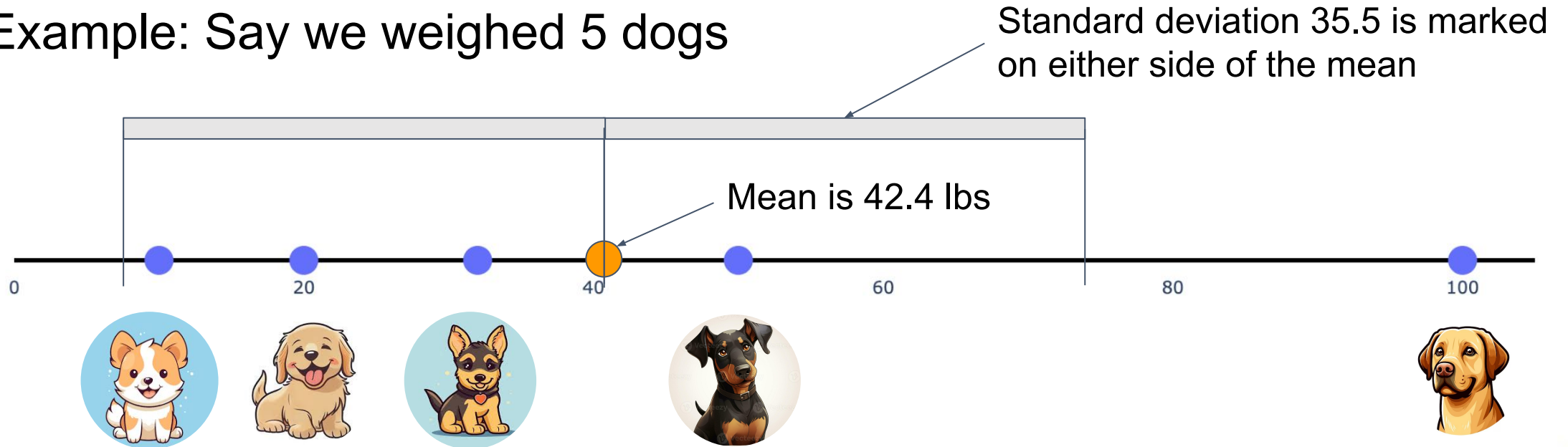
ex. how far a sample mean is likely from the population mean across samples

In other words:

- **Standard Deviation** = variability **within** a sample
- **Standard Error** = estimated variability **across** samples

Standard Error vs Standard Deviation

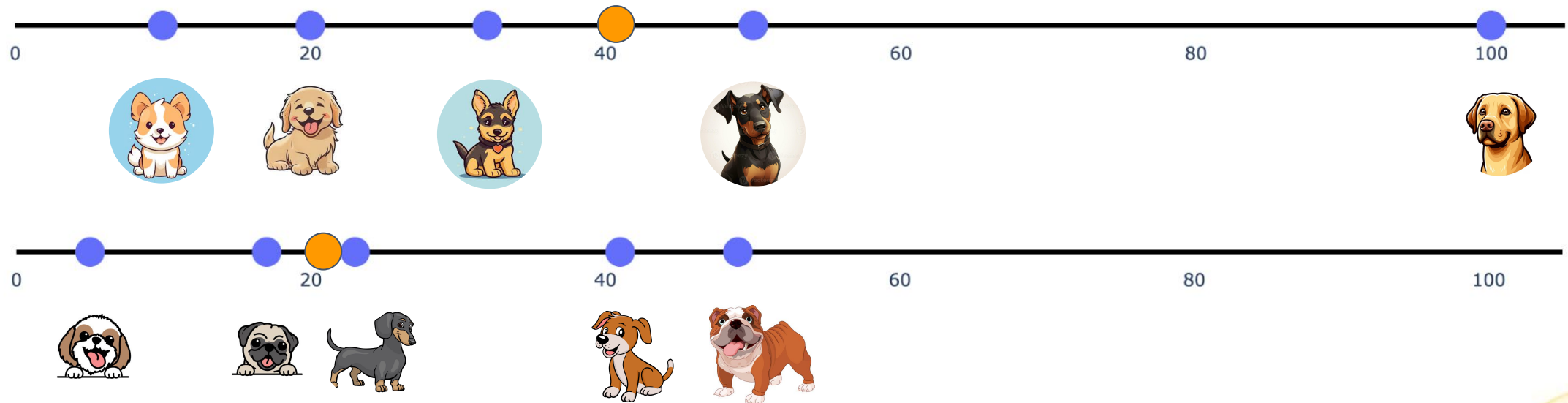
Example: Say we weighed 5 dogs



These measures tell us about how spread out our data is.

Standard Error vs Standard Deviation

Now suppose we did this 5 times with different dogs (25 dogs total)

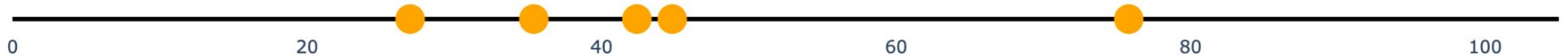


...

We get 5 different sample means, and 5 different sample standard deviations around those means

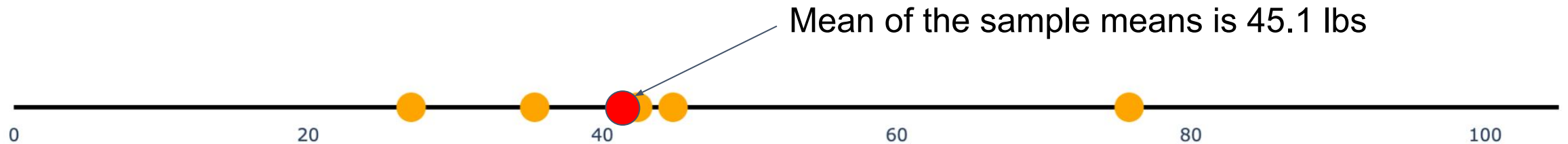
Standard Error vs Standard Deviation

If we plot those 5 sample means on a line, we have



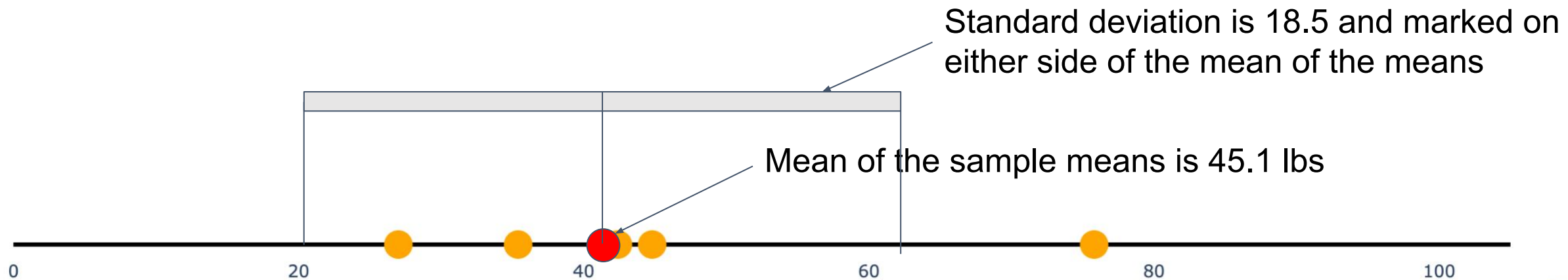
Standard Error vs Standard Deviation

And we can take the mean of those 5 sample means



Standard Error vs Standard Deviation

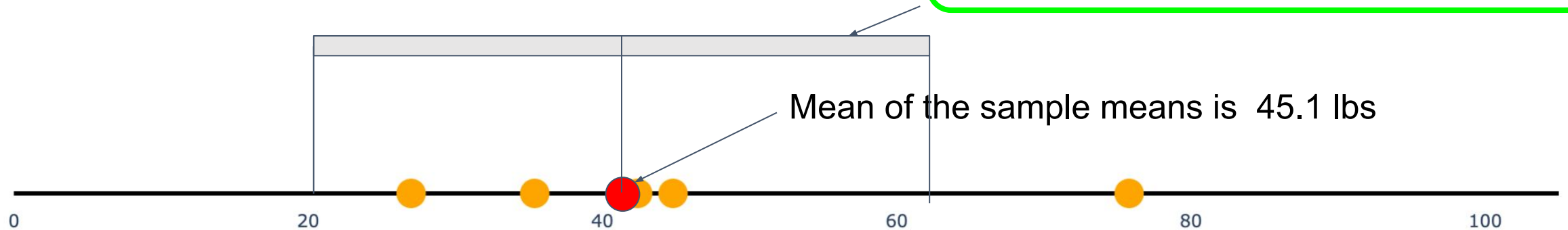
We can calculate the standard deviation of those 5 means



Standard Error vs Standard Deviation

This is the Standard Error

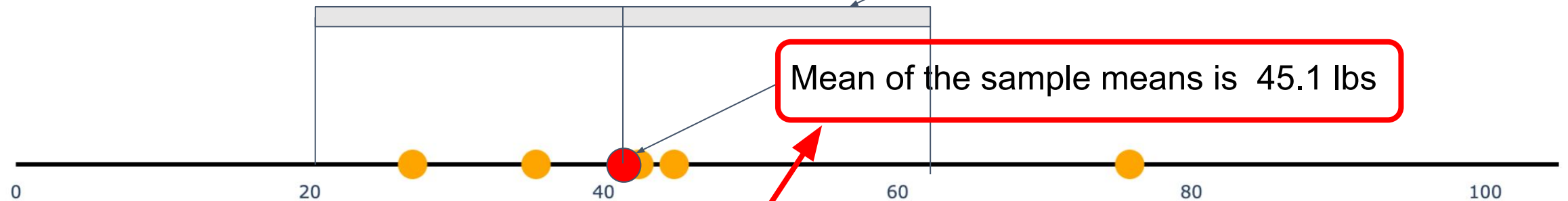
Standard deviation is 18.5 and marked on either side of the mean of the means



Standard Error vs Standard Deviation

If we had infinitely many samples, this would be the Standard Error

Standard deviation is 18.5 and marked on either side of the mean of the means



And recall from CLT that with enough large samples this is a good approximation of the population mean, μ

Standard Error use

You have one sample. The approximated SE tells you how much you would expect the mean to vary if you were to repeat the experiment again, using new samples from the same population.

- Notice that **SE decreases as sample size increases!**
- This makes sense, as bigger samples give us a better approximation of the population



So,

Let's say we're measuring dogs' weight.

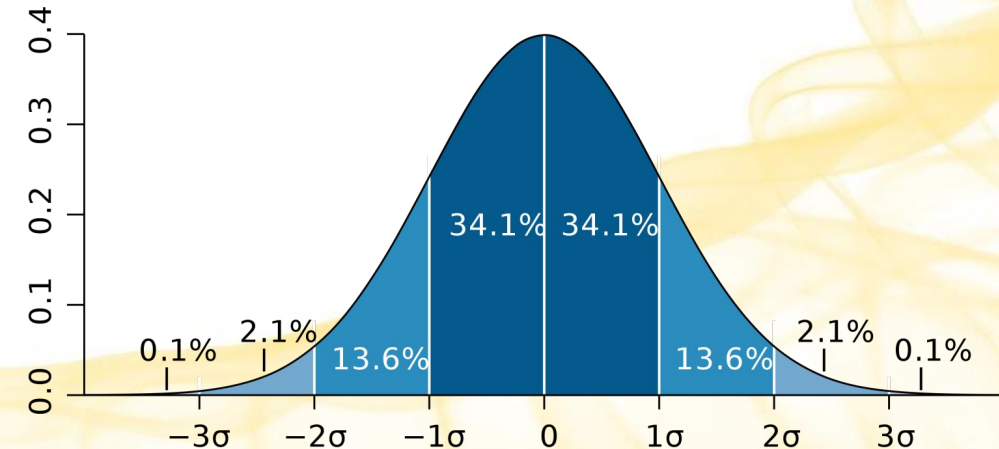
We take 100 samples, each of size 30, and calculate the sample mean (the mean of a sample) for each sample. 100 means total.

The sampling distribution (the distribution of sample means) is normal.

The mean of this normal distribution is equal to the population mean (the average dog weight for all dogs in the population), μ .

The standard deviation of this normal distribution is the Standard Error (by definition of SE).

$$SE = \frac{\sigma}{\sqrt{n}}$$



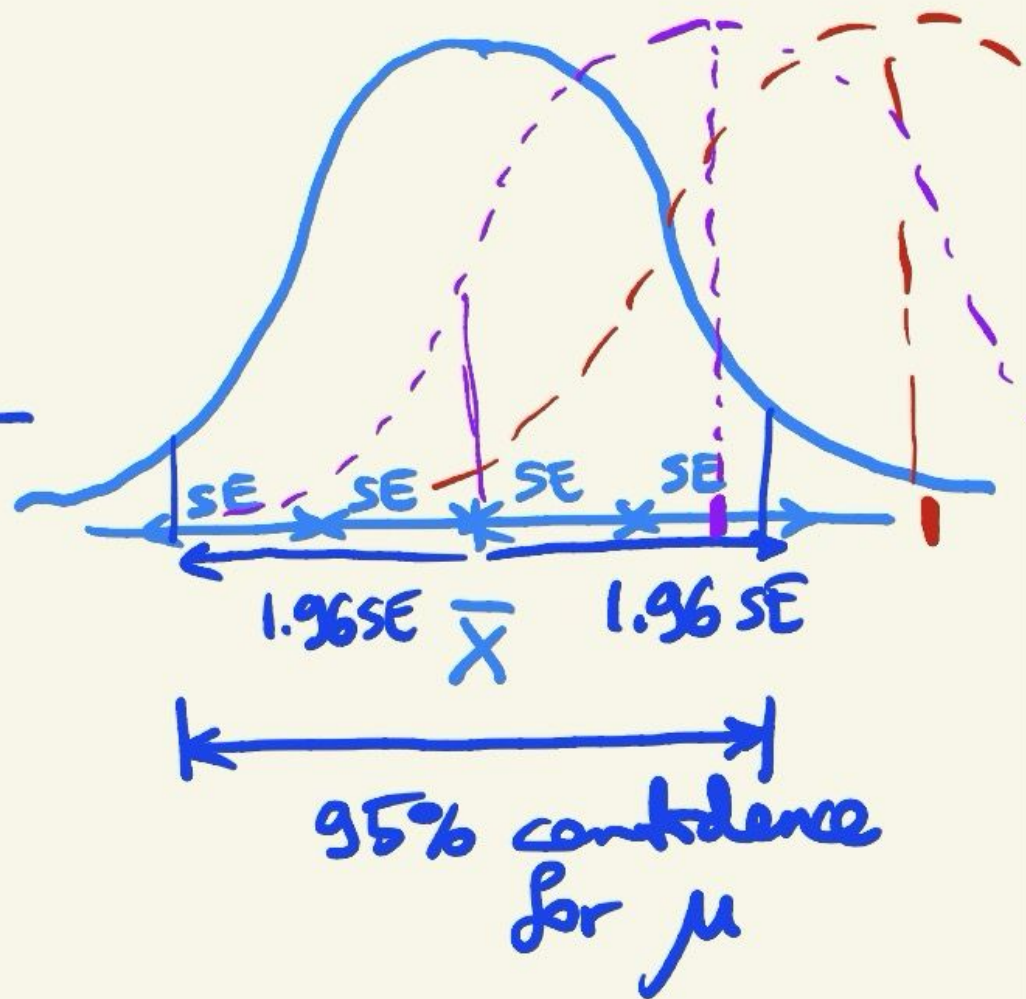
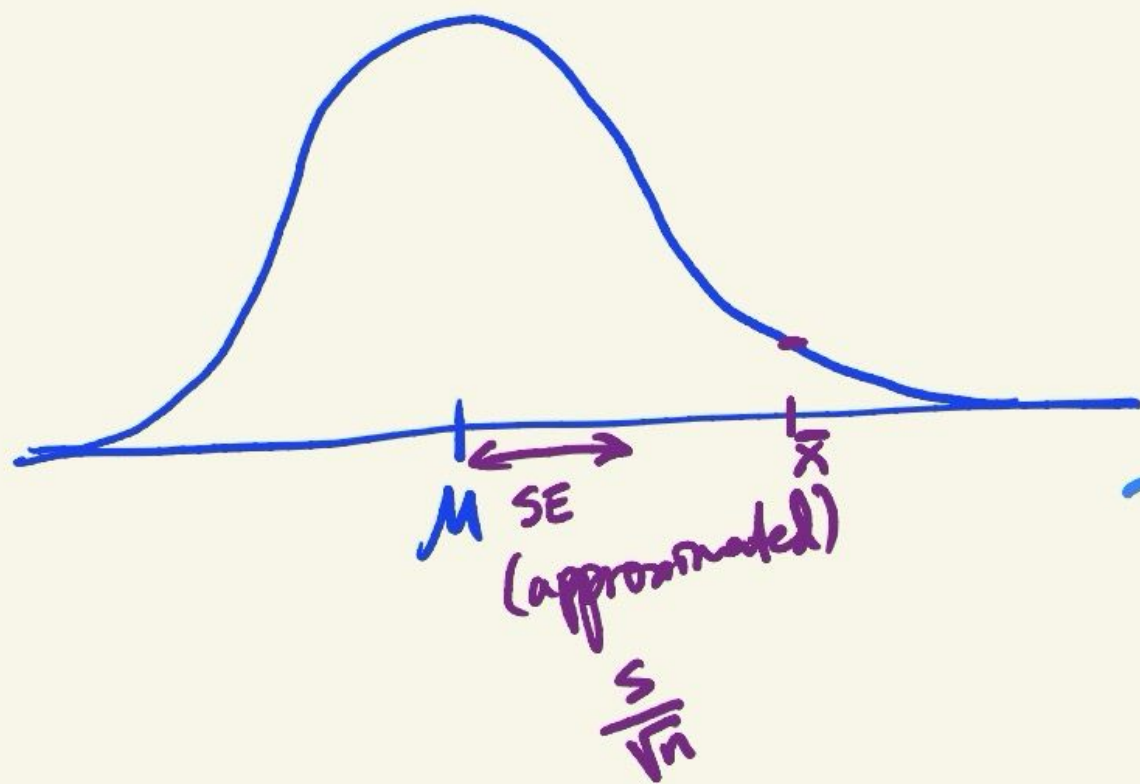
But in reality

We have one sample. You can estimate SE as: $SE \approx \frac{s}{\sqrt{n}}$

What can we say about the population mean from the one sample mean that we did get??

Turning over to the iPad for this (will post the iPad slides)





So,

A 95% confidence interval for the population mean is
sample mean $\pm$ 1.96 (approximated SE)